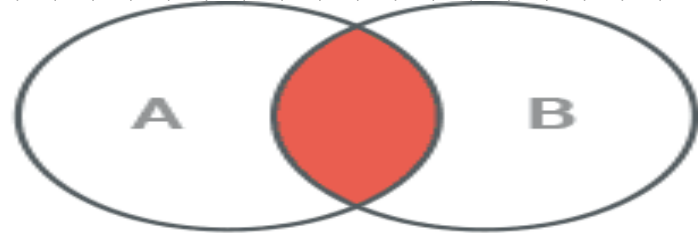
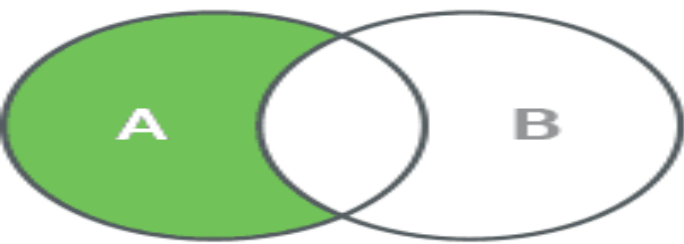


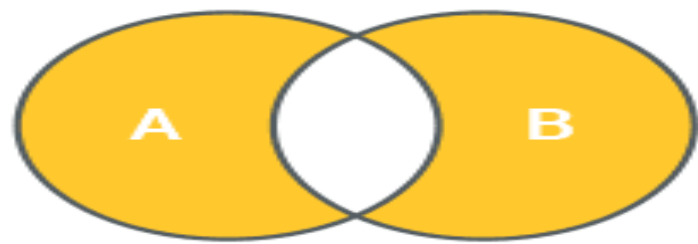
Union



Intersection



Difference



Symmetric Difference

PYnative.com

Set in Python 🐍

$S = \{ 20, 'Jessa', 35.75 \}$

- ✓ **Unordered:** Set doesn't maintain the order of the data insertion.
- ✓ **Unchangeable:** Set are immutable and we can't modify items.
- ✓ **Heterogeneous:** Set can contains data of all types
- ✓ **Unique:** Set doesn't allows duplicates items

frozenset IN PYTHON

What is a frozenset?

- Immutable version of a set
- Unordered
- Contains only unique elements

Methods NOT Supported

- ❌ `fs.add(4)`
- ❌ `fs.remove(2)`
- ❌ `fs.pop()`
- ❌ `fs.clear()`

Real Use Case Example

```
permissions =
frozenset("read, 'write"):
"Editor,
frozenset("read"):
Viewer"
user_access = frozenset
("read, "write")
```

Creating a frozenset

```
fs = frozenset(1, 2, 3, 2)
# Output: frozenset(1, 2, 3)
fs_str = frozenset("hello")
# Output: frozenset("h, 'e, "; 'o)
```

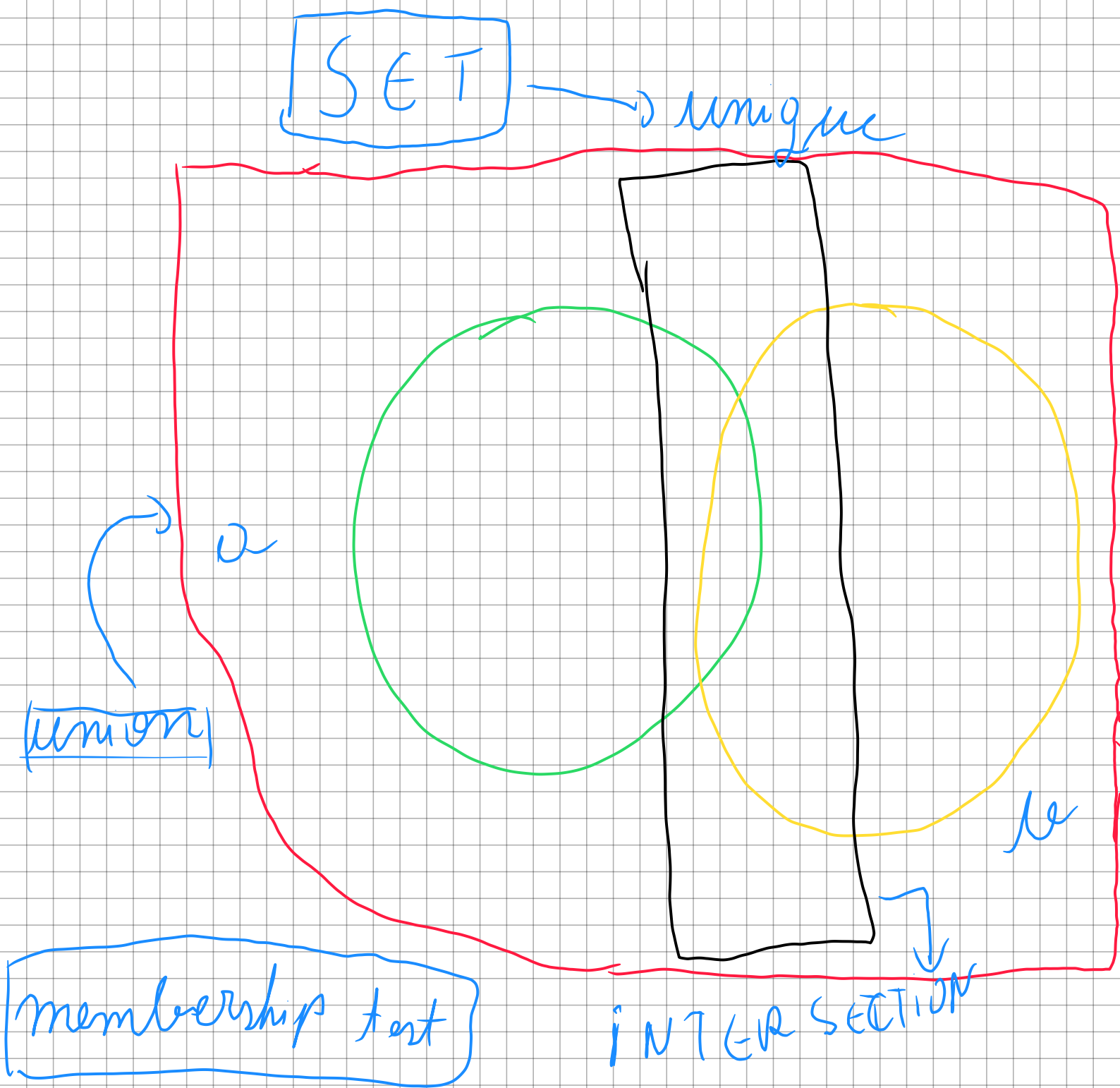
Supported Operations

```
Membership Test    2 in fs
Length            len(fs) 3
Union             a | b# froz-
                  enst(1, 2, 3, 4, 5)
Intersection      a & b# frz
                  enset(5))
Difference        a - b#
                  frozenset(1, 2)
Symmetric         a ^ b#
Difference        enseet (1, 2, 4, 5)
```

Real Use Case Example

▮ Output: Editor

SET AND FROZENSET



essential - spices = { "cardamom", "ginger",
"cinnamon" }

optional - spices = { "cloves", "ginger",
"black pepper" }

all - spices = essential - spices $\cup$ optional -
spices

print (f "All Spices: { all - spices }")

//
{ "ginger", "cloves", "cardamom", "black
pepper", "cinnamon" }

common - spices = essential - spices &
optional - spices

INTERSECTION

print(f"common spices {common - spices}")

//
{ "ginger" }

only - in - essential = essential - spices -
optional - spices

print(f"Only spices: {only - in
- essentials}")

//
{ "common", "cardamom" }

print(f"{'cheves' in essential - spaces? 'cheves' in essential - spaces}")



FALSE

FROZEN SET

SET și FROZENSET în Python

SET

Ce este un set?

Colecție neordonată

NU permite duplicate

Este mutabil

Nu are index

```
s = {1, 2, 3}
```

Atenție:

```
s = {}
```

creează dicționar, nu set.

Set gol corect:

```
s = set()
```

Fără duplicate

```
s = {1, 2, 2, 3}
```

```
print(s)
```

```
# {1, 2, 3}
```

elimină automat duplicatele.

Operații importante

```
s = {1, 2}
```

```
add()
```

```
s.add(3)
```

```
# {1, 2, 3}
```

FROZENSET

Ce este?

Versiune IMUTABILă a setului

Nu poate fi modificat

Poate fi folosit ca cheie în dicționar

Cum se creează

```
fs = frozenset([1, 2, 3])
```

Nu permite modificare

```
fs.add(4)
```

```
# EROARE
```

Permite operații matematice

```
a = frozenset([1, 2])
```

```
b = frozenset([2, 3])
```

```
print(a | b)
```

```
# frozenset({1, 2, 3})
```

remove() s.remove(2)

{1, 3}

union (|)

a = {1, 2}

b = {2, 3}

print(a | b)

{1, 2, 3}

intersection (&)

print(a & b)

{2}

difference (-)

print(a - b)

{1}

Diferen SET vs FROZENSET

CARACTERISTICĂ	SET	FROZEN
MUTABIL	✓	✗
add()	✓	✗
remove()	✓	✗
operatii matematice	✓	✓
POTI FI CHIE IN DICT	✗	✓

Când folosim?

Set când vrem valori unice

Frozenset când vrem valori unice FIXE

Exerci iul 1 – Eliminare duplicate

Ai lista:

```
l = [1, 2, 2, 3, 4, 4, 5]
```

Transform-o într-un set și afi eaz rezultatul.

Rezolvare:

```
l = [1, 2, 2, 3, 4, 4, 5]
```

```
s = set(l)
```

```
print(s)
```

Rezultat:

```
{1, 2, 3, 4, 5}
```

Setul elimină automat duplicatele.

Exerci iul 3 – Frozenset

```
fs = frozenset([10, 20, 30])
```

1

Ce se întâmplă dacă scriem fs.add(40)?

Este permis opera iea fs | {40}?

Rezolvare:

`TypeError` Frozenset este imutabil.

Exerci iul 2 – Opera ii matematice

```
a = {1, 2, 3}
```

```
b = {3, 4, 5}
```

F reuniunea (union)

F intersec ia (intersection)

F diferen a (a - b)

Rezolvare:

```
print(a | b)   # union
```

```
print(a & b)   # intersection
```

```
print(a - b)   # difference
```

Rezultate:

```
{1, 2, 3, 4, 5}
```

```
{3}
```

```
{1, 2}
```

Da, este permis: `print(fs | {40})`

Rezultat:

```
frozenset({10, 20, 30, 40})
```


SET

{ }

For duplicate

Neordonat

Mutable

add()

remove()

union (|)

intersection (&)

difference (-)

FROZENSET

frozenset()

For duplicate

Neordonat

IMUTABIL

Not add()

Not remove()

Permite opera ii matematice